

Notes: Cookson, Lu, Mullins, and Niessner (JFE, 2024)

The Social Signal

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1 Big picture

- **Question.** Are investor social-media platforms interchangeable, or do they aggregate different kinds of market-relevant information? The paper studies whether *attention* and *sentiment* from Twitter, StockTwits, and Seeking Alpha carry common versus platform-specific information, and whether those signals predict returns differently.
- **Setting.** The sample is a firm-day panel from 2012–2021 built from the three major investor social platforms. The focal firms are the 1,500 stocks most discussed on StockTwits. The paper combines social-media measures with RavenPack traditional news, 8-K filings, earnings announcements, CRSP returns, Google search intensity, and retail trading proxies.
- **Core challenge.** Cross-platform correlations could reflect the same underlying news, broad market sentiment, or persistent firm characteristics rather than something genuinely social. Likewise, return predictability could reflect news, overreaction, selection into posting, or measurement differences across platforms.
- **Main conceptual contribution.** The paper separates the “social signal” into two distinct objects:
 - **attention:** which firms people are talking about;
 - **sentiment:** what directional opinions people express about those firms.

It then shows that these two objects behave very differently across platforms and in returns.

- **Main empirical design.** There are three layers of evidence:
 - unconditional and conditional principal components analyses (PCAs) across platforms;
 - return-predictability regressions using common versus platform-specific components of sentiment and attention;
 - two event studies: the May 8, 2019 StockTwits character-limit increase and the January 2021 GameStop (GME) short squeeze with the influx of new retail users.
- **Main baseline result on commonality.** Attention is highly common across platforms, while sentiment is not. In the unconditional PCA, the first attention PC explains **67.0%** of cross-platform attention variation, whereas the first sentiment PC explains only **38.8%**, barely above the 33.3% benchmark from three independent signals.
- **Main robustness result on commonality.** After residualizing signals with respect to traditional news, disclosures, firm fixed effects, and lagged return controls, the pattern barely changes: the first PC of attention still explains about **63.5%** of the variation, while the first PC of sentiment still explains only about **38.1%**.
- **Main user-composition result.** Within StockTwits itself, attention is highly common across investor types (top 1% by followers, professionals, intermediates, novices, and users with no label), with PC1 explaining **84.3%**. Sentiment across those user groups is much less common, with PC1 explaining only **27.6%**. So weak sentiment commonality is not just an NLP artifact across platforms.
- **Main return result.** The common sentiment component predicts **positive next-day abnormal returns**, while the common attention component predicts **negative next-day abnormal**

returns. In Table 6, a one-standard-deviation increase in sentiment PC1 predicts about **+6.1 bps** on day $t+1$, while a one-standard-deviation increase in attention PC1 predicts about **−13.1 bps**, conditional on sentiment.

- **Main dynamic result.** Sentiment’s positive return effect does *not* reverse over the following 20 days, which is consistent with information. Attention behaves differently: it is associated with strongly positive same-day returns and retail buying, followed by a *partial reversal* over roughly 10–20 days, which is consistent with overreaction on high-attention days.
- **Main platform-specific result.** StockTwits on its own most closely reflects the common social signal. In one-platform regressions, StockTwits sentiment and attention have coefficient signs and magnitudes close to the common PC1s, whereas Twitter attention is largely offset by an idiosyncratic positive component and Seeking Alpha attention does not appear to load on the common attention factor.
- **Main feature-shock result.** After StockTwits increased its character limit from 140 to 1,000 characters, StockTwits sentiment became more informative for next-day returns. The increase is strongest for stocks whose messages are in the top quartile of daily average length. Twitter and Seeking Alpha do not show a comparable break.
- **Main user-inflow result.** After the GME event, the informativeness of sentiment falls sharply. The decline is concentrated among *new* StockTwits users rather than established users, and coincides with a large surge in “short squeeze” discussion among the new-user group.
- **Economic takeaway.** Investor social media are not one homogeneous dataset. Attention is a common, platform-spanning object and seems tied to retail overreaction; sentiment is much more platform- and user-specific and appears to contain return-relevant information. Platform design and user composition materially shape what the “social signal” means.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- The paper compares the three most established investor social-media platforms rather than studying only one: **StockTwits**, **Twitter**, and **Seeking Alpha**.
- These platforms differ sharply in message length, moderation, topic focus, and user base. Those differences are precisely what the paper wants to map into differences in the information content of attention and sentiment.
- The 2012–2021 window contains both “normal” years and two natural events that are useful for interpretation: the StockTwits character-limit change and the GME/retail-trading boom.

Interpretation. This is a good laboratory because the paper can compare multiple social-media ecosystems at the same firm-day frequency while holding the underlying stock market fixed.

2.2 Data source and sample construction

Twitter.

- Data come from Social Market Analytics (SMA).
- The paper uses daily 4:00 p.m. snapshots of financially oriented tweets: the number of tweets and average sentiment over the previous 24 hours for each firm.

Seeking Alpha.

- Article-level sentiment comes from RavenPack 1.0.
- The sample keeps only articles with relevance score above 75.
- Article sentiment is RavenPack’s Event Sentiment Score (ESS), which lies in $[-1, 1]$.

StockTwits.

- The raw data are message-level and comprehensive.
- The paper keeps all single-firm messages from 2010–2021: about **150 million** messages from more than **800,000** users.
- Users posting more than 1,000 tweets in a day are dropped.
- The paper restricts the firm universe to the **top 1,500** firms by StockTwits message count over 2010–2021.

Final analysis sample.

- The analysis begins in 2012 because Twitter and Seeking Alpha data are sparse before then.
- Full merged social-media panel before the “10 StockTwits messages” restriction: about **2.80 million** firm-days.
- Requiring at least 10 StockTwits messages reduces this to about **821,534** firm-days.
- Adding controls yields roughly **815,980** firm-days; adding returns yields roughly **814,646** firm-days.

Interpretation. The sample is built around days with enough StockTwits traffic to measure the “social signal” cleanly, while still linking that activity to Twitter and Seeking Alpha.

2.3 Defining sentiment and attention

For StockTwits and Seeking Alpha, firm-day sentiment is the average sentiment over all messages or articles about firm i from the market close on day $t - 1$ to the market close on day t . Twitter sentiment is already provided at that firm-day level by SMA.

For StockTwits, the paper observes two sentiment measures:

- **self-labeled sentiment:** users can explicitly tag a post as bullish or bearish, coded as +1 or −1;
- **MarketLex sentiment:** a proprietary NLP score in $[-1, 1]$.

The key attention measure is the firm’s share of total platform message volume that day:

$$\text{Attention}_{i,t} = \frac{\text{Messages}_{i,t}}{\sum_i \text{Messages}_{i,t}}. \quad (1)$$

Interpretation.

- Sentiment captures *direction* of views.
- Attention captures *allocation of discussion* across firms, not raw message counts alone.

The paper also checks robustness to two alternatives:

- “abnormal attention” = deviation of message count from its median over the prior 10 days;
- an extensive-margin attention measure equal to whether a stock is mentioned on a platform that day.

2.4 Traditional news, returns, and retail-trading data

Traditional news controls.

- Dow Jones Newswire (DJNW) and Wall Street Journal (WSJ) coverage and sentiment from RavenPack 1.0.
- 8-K filing dates from SEC Analytics Suite.
- Earnings announcement dates from IBES.

Returns data.

- Daily abnormal returns are computed from CRSP by subtracting the value-weighted market return from the stock’s return.

Retail-trading outcomes.

- Retail trade imbalance comes from the Boehmer et al. (2021) style measure.
- Robinhood user ratio follows Barber et al. (2022): user count around 4 p.m. on day t relative to the user count around 4 p.m. on day $t - 1$.

Other controls.

- Google attention is measured using log abnormal Google Search Volume Index (ASVI), following Niessner (2015).
- Return controls include lagged volatility and lagged cumulative abnormal returns.

2.5 Platform features and user groups

The paper emphasizes that social-media platforms are *not* mechanically comparable.

- **Seeking Alpha** is long-form and lightly moderated.
- **Twitter** is a general-purpose platform with short posts and threadability.
- **StockTwits** is finance-specific, uses cashtags, allows bullish/bearish self-labeling, and had a hard character cap of 140 before May 8, 2019 and 1,000 afterwards.

The StockTwits data also allow the paper to split signals by user type:

- influencers (top 1% by followers),
- professionals,
- intermediates,
- novices,
- users with no experience label.

For the GME analysis, StockTwits users are further split into:

- **old users:** joined before January 1, 2020;
- **new users:** joined in or after 2020.

Interpretation. These user partitions are crucial because they help distinguish between platform-design explanations and user-composition explanations.

2.6 Dependent variables

The paper uses four main classes of dependent variables.

- **Cross-platform commonality:** PC loadings and fraction of explained variation from PCAs of attention and sentiment.
- **Next-day abnormal returns:** $AR_{i,t+1}$.
- **Dynamic cumulative abnormal returns:** cumulative abnormal return from day $t+1$ through horizons up to $t + 20$.
- **Same-day retail trading:** retail trade imbalance and Robinhood user-ratio outcomes.

2.7 Descriptive facts

- StockTwits is much denser than the other platforms: the average firm-day has about **132.4** StockTwits messages, versus **18.84** tweets on Twitter and only **0.46** Seeking Alpha articles.
- Mean firm-day sentiment is positive on all three platforms, with StockTwits highest at about **0.10** and Twitter/Seeking Alpha around **0.02**.
- StockTwits is small-cap oriented in coverage: about **68.6%** of the top 1,500 covered firms are small-cap, compared with about **30%** on Twitter and Seeking Alpha.
- Even so, large firms attract a large share of total messages on Twitter and Seeking Alpha: around **61.7%** and **57.1%** of message volume respectively.

Interpretation. The platforms differ not only in volume, but also in who and what they pay attention to. That heterogeneity becomes especially important for sentiment.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper's underlying logic is simple:

- if many investors across platforms are talking about the same firms, attention should look common;
- if platforms aggregate different user groups, conversation styles, and communication constraints, sentiment can easily look idiosyncratic even for the same firm-day.

The return hypotheses are also distinct:

- **sentiment** may forecast returns if it contains return-relevant information;
- **attention** may forecast negative next-day returns if it is associated with retail overreaction and later partial reversal.

The event-study hypotheses sharpen interpretation:

- a platform-design shock (the StockTwits character-limit increase) should change informativeness if communication constraints matter;
- a user-composition shock (the GME event and the influx of new retail users) should change informativeness if who is posting matters.

3.2 Unconditional cross-platform PCA

The paper first performs PCAs on six firm-day signals: attention and sentiment for each of the three platforms.

The key output is not just the loadings, but also the share of total variation explained by each PC.

- If PC1 loads mainly on attention signals and explains a large share of variance, attention is common across platforms.
- If PC1 for sentiment explains little more than the independent benchmark, sentiment is mostly platform-specific.

Main unconditional findings.

- Panel A of Table 2 shows that attention is much more correlated than sentiment. For StockTwits, the correlation with Twitter is about **0.595** for attention but only **0.125** for sentiment; with Seeking Alpha it is **0.398** for attention and only **0.038** for sentiment.
- In the six-signal PCA, PC1 explains **35.6%** of total variation and loads primarily on attention. PC2 explains **19.3%** and loads primarily on sentiment.
- When the PCA is run separately by signal type, attention PC1 explains **67.0%**, while sentiment PC1 explains only **38.8%**.

Interpretation. Attention behaves like a common social “topic” signal. Sentiment behaves much more like a platform- or user-specific opinion signal.

3.3 Conditional PCA: removing news and firm-level confounds

To purge common variation coming from conventional news or fixed firm traits, each signal is residualized according to:

$$\text{Signal}_{i,t}^P = \Gamma^P X_{i,t} + \gamma_i^P + \varepsilon_{i,t}^P, \quad (2)$$

where P indexes platform, $X_{i,t}$ includes traditional-news controls and event controls, and γ_i^P are firm fixed effects.

The paper implements three versions of this residualization:

- residualize by news only;
- residualize by news and firm fixed effects;
- residualize by news, firm fixed effects, lagged volatility, and lagged cumulative abnormal returns.

Main conditional findings.

- Residualized attention PC1 still explains about **66.4%**, **63.4%**, and **63.5%** of variation across the three residualization schemes.
- Residualized sentiment PC1 remains around **38.2%**, **38.2%**, and **38.1%**.

Interpretation. The gap between common attention and idiosyncratic sentiment is *not* just a reflection of news coverage, firm disclosures, persistent firm heterogeneity, or recent return dynamics.

3.4 Heterogeneity by firm size

Because StockTwits covers relatively more small firms while Twitter and Seeking Alpha skew toward larger firms, the paper repeats the PCA within three market-cap bins:

- **small:** below \$2 billion,
- **medium:** \$2–10 billion,
- **large:** above \$10 billion.

Main findings from Table 4.

- Attention commonality rises strongly with firm size: attention PC1 explains **49.5%** for small firms, **58.9%** for medium firms, and **72.5%** for large firms.
- Sentiment commonality also rises, but modestly: sentiment PC1 explains **36.3%**, **40.1%**, and **42.4%** across the three bins.
- After residualization, the same pattern remains.

Interpretation. Large-cap names are “common knowledge” across platforms to a much greater extent than small-cap names. That matters for external validity: sentiment-based results from one platform may generalize least well exactly where retail-social-media activity is strongest—small caps.

3.5 User-type PCAs inside StockTwits

The paper then holds platform and NLP fixed by staying inside StockTwits and comparing user groups.

Main findings from Table 5.

- Attention correlations between each user group and its complement are very high: **0.805** for top 1% users, **0.860** for professionals, **0.948** for intermediates, **0.914** for novices, and **0.947** for no-label users.
- Sentiment correlations are much lower: from **0.088** to **0.166**.
- Residualized attention PC1 across user groups explains **84.3%** of variation.
- Residualized sentiment PC1 explains only **27.6%**.

Interpretation. The paper’s central “attention common, sentiment fragmented” result survives even when platform design and NLP are held fixed. So differences in *who posts* are themselves enough to generate fragmented sentiment.

3.6 Persistence of signals

The paper computes partial autocorrelations out to 20 lags (days).

- Attention is much more persistent: lag-1 autocorrelation is around **0.8** and decays only after about five days.
- Sentiment is much less persistent: lag-1 autocorrelation lies roughly between **0.1** and **0.25** and decays more quickly.

Interpretation. This motivates the return regressions including ten lags of both sentiment and attention. Without those controls, attention in particular would be mechanically picking up its own strong serial dependence.

3.7 Return-predictability regression

The core return specification is

$$AR_{i,t+\tau} = \beta_1 \text{Attention}_{i,t} + \beta_2 \text{Sentiment}_{i,t} + X_{i,t}\Gamma + \alpha_i + \alpha_t + \varepsilon_{i,t}, \quad (3)$$

where $\tau \in \{0, 1, \dots, 20\}$.

Controls in $X_{i,t}$.

- DJNW sentiment and attention,
- 8-K and earnings-announcement indicators,
- lagged volatility over days $t - 1$ to $t - 5$,
- lagged cumulative abnormal returns over $(t - 1, t - 5)$ and $(t - 6, t - 30)$,
- log abnormal Google ASVI,

- ten lags of sentiment and attention.

Interpretation of coefficients.

- Regressions with only one platform’s signal mix common and idiosyncratic components.
- Regressions with all platforms separately isolate the platform-specific component, because each platform is conditioned on the others.
- Regressions with PC1s target the common component directly.

Main findings from Table 6.

- Sentiment PC1 predicts **+0.061%** next-day abnormal return per standard deviation.
- Attention PC1 predicts **−0.131%** next-day abnormal return per standard deviation, conditional on sentiment.
- StockTwits on its own closely matches the common signal: sentiment **+0.053%**, attention **−0.151%**.
- In the multivariate all-platform regression, Twitter attention has a **positive** idiosyncratic coefficient (about **+0.112%**), but in the one-platform regression its combined effect is near zero. That means Twitter’s common and idiosyncratic attention components point in opposite directions.
- Seeking Alpha sentiment is strongly positive in both the all-platform and one-platform regressions (about **+0.079%** to **+0.081%**), while Seeking Alpha attention is close to zero or slightly negative.

3.8 Dynamic returns and retail-trading regressions

The paper next replaces $AR_{i,t+1}$ with cumulative abnormal returns from $t + 1$ through horizons up to $t + 20$, and separately studies same-day retail order imbalance and Robinhood-user changes.

Main findings.

- Sentiment’s positive return effect persists out to day $t + 20$ with *no reversal*.
- High-attention days exhibit strong positive same-day returns, followed by a gradual and partial reversal over the next 10–20 days.
- Same-day retail buying is positively related to both sentiment and attention, especially attention.
- The retail-buying relation decays quickly after day t , so the multi-day attention reversal is not simply a story of persistent retail pressure.

Interpretation. The paper’s preferred reading is:

- sentiment contains genuine return-relevant information;
- attention captures investor salience and contemporaneous retail demand that overshoots and then partially reverses.

3.9 Event-study specifications

To study whether platform design or user composition changes the signal, the paper estimates

$$AR_{i,t+1} = \beta_1 \text{Attention}_{i,t} + \beta_2 \text{Sentiment}_{i,t} + \beta_3 \text{Sentiment}_{i,t} \times \text{Post}_t + \beta_4 \text{Attention}_{i,t} \times \text{Post}_t + X_{i,t} \Gamma + \alpha_i + \alpha_t + \varepsilon_{i,t}. \quad (4)$$

Two events define Post_t :

- **StockTwits character-limit event:** post May 8, 2019.
- **GameStop event:** post February 1, 2021, with January 2021 omitted.

The paper also extends Eq. (4) with treatment interactions:

- treated = stocks whose messages are in the top quartile of average message length (for the character-limit analysis);
- treated = new StockTwits users (for the GME/new-user analysis).

Interpretation. These event studies are not broad claims that social media causally moves prices. They are narrower: they ask whether the *informativeness* of the signal changes when platform constraints or user mix change.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimates

- In the PCA analysis, identification comes from cross-platform comovement of firm-day signals.
- In the return regressions, firm fixed effects absorb time-invariant differences across firms, and date fixed effects absorb market-wide shocks on each day.
- The coefficients are thus identified from within-firm variation over time and cross-sectional dispersion across firms on the same day, conditional on rich news and lagged-return controls.

Interpretation. The baseline return results are best read as *predictive informativeness* results, not as clean causal treatment effects of social media on returns.

4.2 Why reverse causality is limited, but not fully eliminated

The main timing choice is that social signals are measured on day t , and the main outcome is abnormal return on day $t + 1$.

That helps because:

- same-day return news is not mechanically in the dependent variable;
- lagged cumulative returns and lagged volatility are controlled for;
- ten lags of sentiment and attention soak up persistence in the social signals.

Still, the paper does *not* claim to have an instrument that fully isolates exogenous day- t sentiment or attention. It therefore interprets the baseline coefficients as evidence about information content, not as a structural price-impact estimate.

4.3 Traditional-media and firm-level confounds

The conditional PCA is the main device that makes the baseline commonality results credible.

- RavenPack news sentiment and coverage are residualized out.
- 8-K and earnings-announcement timing is controlled for.
- Firm fixed effects remove permanent platform-firm match effects.
- Additional residualization by lagged return volatility and lagged cumulative abnormal returns shows the same pattern.

Bottom line. If common attention were merely the shadow of conventional news, or if fragmented sentiment came only from persistent firm-level differences, the gap between attention and sentiment PC1s should collapse after residualization. It does not.

4.4 Why the paper does not rely on NLP differences alone

A natural concern is that different sentiment algorithms across platforms produce artificial fragmentation in sentiment.

The within-StockTwits user-type PCA is the paper’s strongest answer:

- the platform is held fixed;
- the NLP is held fixed;
- attention is still common and sentiment is still weakly common.

The paper also reruns the character-limit analysis using only self-labeled StockTwits sentiment. The effect remains similar, suggesting the character-limit result is not simply “longer text improves NLP classification.”

4.5 Firm-size composition is not the whole story

Because StockTwits overrepresents small-cap firms, one could worry that the core cross-platform results are a size-composition artifact. Table 4 addresses this directly.

- The “attention more common than sentiment” result holds within small, medium, and large stocks.
- Large firms do show more commonality, but the qualitative ranking does not flip.

Interpretation. The paper’s main result is not “just” a small-cap phenomenon, even though small-cap stocks are the part of the market where platform differences matter most.

4.6 What the event evidence adds

The paper has no classical instrumental variable for baseline social-media signals. Instead, it uses two quasi-experimental shocks to sharpen interpretation.

1. StockTwits character-limit increase.

- This is a within-platform design shock.
- Only StockTwits sentiment becomes more informative afterwards.
- The increase is strongest exactly where the feature should matter most: long messages.
- Event-study coefficients show no obvious pre-trend.

2. GME plus new-user influx.

- Informativeness of sentiment falls sharply after the event.
- The decline is concentrated among users who joined recently, not established users.
- “Short squeeze” discussion rises disproportionately for the new-user group.

Interpretation. Together, these events make it credible that both platform design and user composition are economically important determinants of what a platform-level social signal means.

4.7 What the paper can and cannot distinguish

What it establishes fairly convincingly.

- Attention is much more common across platforms than sentiment.
- This difference survives conventional-news controls, firm fixed effects, and within-StockTwits user-type comparisons.
- Sentiment and attention have different return signatures.
- Platform design and user composition both shift informativeness.

What it does not fully pin down.

- It does not cleanly separate *naive extrapolation* from *selective communication*; the positive-return/no-reversal pattern for sentiment and the positive-same-day/negative-next-day pattern for attention are consistent with those stories, but not uniquely identified by them.
- It does not provide a full structural model of how information is produced and transmitted across platforms.
- It does not claim that the baseline coefficients are a causal price-impact estimate of social-media posts.

5 Tables and figures

5.1 Table 1 and Figure 2: what the platforms cover and how similar they are

Table 1

Summary statistics.

Panel A: Statistics by social media platform												
	Daily sentiment				# messages (daily)				# of firms			Firm-day observations
	Mean	S.D.	Min	Max	Mean	S.D.	Min	Max	Ever mentioned	All	Mentioned	All
StockTwits	0.10	0.14	-0.97	0.97	132.40	734.97	10	138,280	1,497	1,500	815,980	815,980
Twitter	0.02	0.06	-0.80	0.94	18.84	62.69	0	7,160	1,271	1,500	522,284	815,980
Seeking Alpha	-0.02	0.12	-1	1	0.46	1.75	0	150	1,283	1,500	137,018	815,980

Panel B: Statistics by user type on StockTwits												
	Daily sentiment				# messages (daily)				Users		Firm-day observations	
	Mean	S.D.	Min	Max	Mean	S.D.	Min	Max	#	Share	Non-zero	All
Top 1%	0.07	0.29	-1	1	5.97	34.04	0	4,212	7,173	0.01	512,549	815,980
Professional	0.09	0.30	-1	1	7.89	28.92	0	2,405	20,073	0.02	591,383	815,980
Intermediate	0.09	0.29	-1	1	13.37	50.88	0	5,439	45,156	0.05	687,993	815,980
Novice	0.07	0.29	-1	1	5.87	26.18	0	3,645	34,118	0.04	514,773	815,980
No label	0.10	0.18	-0.99	0.99	105.27	658.60	0	127,243	730,154	0.88	810,614	815,980

Panel C: Stock characteristics by social media platform												
	Share of firms (%)			Share of messages (%)								
	StockTwits	Twitter	Seeking Alpha	StockTwits	Twitter	Seeking Alpha						
Small-cap	68.60	30.07	30.67	54.27	16.13	18.27						
Mid-cap	15.13	36.20	37.27	15.41	22.21	24.65						
Large-cap	16.27	33.73	32.07	30.32	61.66	57.09						

Panel D: Firm-day observations satisfying sample restriction												
Sample restriction						# obs.			# dropped obs.			
Full sample						2,795,852			-			
At least 10 StockTwits messages						821,534			1,974,318			
Non-missing controls data						815,980			5,354			
Non-missing controls + returns						814,646			1,334			

Note: Panel A reports statistics on the firm-day level sentiment and attention by social media platform for all observations with at least 10 StockTwits messages. The sample time frame is Jan. 1, 2012 to Dec. 31, 2021 for StockTwits, Twitter, and Seeking Alpha. "# of firms - Ever mentioned" refers to the # of firms ever mentioned on a platform during our sample period. "# of firms - All" refers to the # of firms included in our analysis sample (with the sentiment of firms not mentioned replaced by zeros). "Firm-day observations - Mentioned" refers to the # of firm-day observations with non-zero attention; "firm-day observations - All" refers to the # of firm-day observations in our analysis sample. Panel B provides similar statistics by user type on StockTwits. "Users - # (or Share)" refers to the # (or share) of StockTwits users of a certain type. Panel C reports the share of the 1,500 most talked-about firms on each platform that are in each of three market capitalization bins (first three columns), and the share of messages about firms in each bin (columns 4-6). "Small-cap," "mid-cap," and "large-cap" refer to stocks with market capitalizations below 2 billion, between 2 and 10 billion, and above 10 billion. Panel D shows how sample restrictions reduce the # of firm-day observations to arrive at our analysis sample.

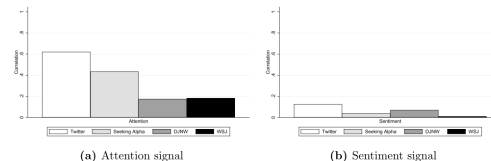


Fig. 2. Cross-platform correlation of social signals. Note: This figure reports the bivariate correlations of attention and sentiment signals between StockTwits and other platforms or news sources at the firm-day level. Attention is measured as the fraction of messages, reports, or articles about a firm across all firms on a platform in a day. Sentiment is measured as the average sentiment of all messages about a firm on a platform in a day.

Read:

- Table 1 shows that StockTwits is far denser than the other platforms: 132.4 average daily messages per firm-day versus 18.84 for Twitter and 0.46 for Seeking Alpha.
- Table 1 also shows that StockTwits covers far more small-cap names than Twitter or Seeking Alpha.
- Figure 2 shows the central descriptive contrast: attention is strongly correlated across platforms, while sentiment is weakly correlated.
- On the figure, StockTwits attention correlates with Twitter at roughly 0.6 and with Seeking Alpha around 0.4, but sentiment correlations are close to zero.

Economic message. The three platforms talk about many of the same *stocks*, but they do not express the same *opinions* about those stocks. That distinction drives the whole paper.

5.2 Table 2 and Table 3: unconditional versus conditional PCA

Read:

- Table 2 shows that attention PC1 explains 67.0% of cross-platform variation, versus only 38.8% for sentiment PC1.
- Table 3 shows that after residualizing news, firm fixed effects, and lagged market variables, attention PC1 still explains about 63.5% while sentiment PC1 stays at about 38.1%.
- The gap therefore survives the paper's main confounding controls.

Economic message. This is the strongest evidence that the “attention common / sentiment fragmented” result is structural to the social-media environment rather than a by-product of traditional news.

Table 2

How common is sentiment and attention across platforms?

Panel A: Correlations with the StockTwits signal						
	Twitter	Seeking Alpha	DJNW	WSJ		
StockTwits attention	0.595	0.398	0.220	0.163		
StockTwits sentiment	0.125	0.038	0.032	0.010		
Panel B: PCA of all platform-level signals						
	PC1	PC2	PC3	PC4	PC5	PC6
Attention:						
StockTwits	0.548	-0.130	-0.188	0.047	0.638	0.488
	(0.052)	(0.107)	(0.292)	(0.069)	(0.155)	(0.076)
Twitter	0.605	-0.033	-0.098	-0.009	0.048	-0.788
	(0.013)	(0.050)	(0.126)	(0.016)	(0.142)	(0.025)
Seeking Alpha	0.548	-0.007	0.052	0.084	-0.745	0.368
	(0.017)	(0.047)	(0.180)	(0.021)	(0.212)	(0.117)
Sentiment:						
StockTwits	-0.031	0.644	-0.345	0.682	0.017	-0.014
	(0.014)	(0.011)	(0.337)	(0.010)	(0.056)	(0.006)
Twitter	0.082	0.647	-0.225	-0.720	-0.008	0.071
	(0.046)	(0.041)	(0.282)	(0.013)	(0.035)	(0.024)
Seeking Alpha	0.160	0.384	0.885	0.087	0.190	0.008
	(0.070)	(0.021)	(0.737)	(0.040)	(0.221)	(0.051)
Fraction	35.6%	19.3%	15.9%	14.5%	9.2%	5.5%
	(4.284)	(0.124)	(0.482)	(0.085)	(2.240)	(1.638)
Panel C: PCA of platform-level attention or sentiment signals						
	Attention signal			Sentiment signal		
	PC1	PC2	PC3	PC1	PC2	PC3
StockTwits	0.565	-0.665	0.489	0.611	-0.464	0.642
	(0.027)	(0.592)	(0.073)	(0.006)	(0.021)	(0.010)
Twitter	0.614	-0.057	-0.787	0.662	-0.147	-0.735
	(0.029)	(0.075)	(0.020)	(0.003)	(0.015)	(0.002)
Seeking Alpha	0.551	0.745	0.376	0.435	0.874	0.217
	(0.022)	(0.620)	(0.109)	(0.009)	(0.009)	(0.024)
Fraction	67%	18.9%	11.1%	38.8%	32.3%	29%
	(9.336)	(6.033)	(3.360)	(0.170)	(0.069)	(0.144)

Note: This table reports the correlations and principal component analyses of social signals across platforms. Panel A reports the bivariate correlations of attention and sentiment between StockTwits and another platform. Panel B reports the principal components of attention and sentiment signals in one analysis, while panel C reports the principal components separately for attention (columns 1–3) and for sentiment (columns 4–6). Standard errors in parentheses are double clustered by firm and by date via a block bootstrap procedure following [Thompson \(2011\)](#).

Table 3
How common is the social signal across platforms? Conditional on news and firm fixed effects.

Panel A: PCA of residualized attention signals									
	Residualize news			Residualize news & firm			Residualize news, firm, & returns		
	PC1	PC2	PC3	PC1	PC2	PC3	PC1	PC2	PC3
StockTwits	0.582	-0.548	0.601	0.608	-0.421	0.674	0.606	-0.427	0.671
Twitter	0.616	-0.185	-0.766	0.627	-0.266	-0.732	0.626	-0.266	-0.733
Seeking Alpha	0.531	0.815	0.230	0.497	0.867	0.183	0.492	0.864	0.186
Fraction	66.4%	21%	12.7%	62.4%	24.1%	12.5%	63.5%	23.9%	12.6%
	(8.768)	(5.380)	(3.666)	(7.898)	(4.449)	(3.536)	(8.151)	(4.513)	(3.716)
Panel B: PCA of residualized sentiment signals									
	Residualize news			Residualize news & firm			Residualize news, firm, & returns		
	PC1	PC2	PC3	PC1	PC2	PC3	PC1	PC2	PC3
StockTwits	0.639	-0.375	0.671	0.660	-0.288	0.694	0.660	-0.289	0.694
Twitter	0.675	-0.144	-0.724	0.676	-0.174	-0.716	0.676	-0.173	-0.716
Seeking Alpha	0.369	0.915	0.162	0.327	0.942	0.080	0.327	0.942	0.081
Fraction	38.2%	32.7%	29.2%	38.2%	32.8%	29.1%	38.1%	32.8%	29.2%
	(8.195)	(6.664)	(6.143)	(8.134)	(6.698)	(6.135)	(8.134)	(6.698)	(6.135)

Note: This table repeats the principal component analysis in Table 2 using the residualized social signal. The residualized signal in columns 1–3 refers to the residual from regressing a signal on DJNW sentiment (lagged 0 through 7 days), DJNW attention (lagged 0 through 7 days), dummies for earnings announcements (lagged 0 through 7 days), dummies for 8-K filings (lagged 0 through 7 days). In columns 4–6, we also residualize out firm fixed effects. In columns 7–9, we further residualize out lagged return volatility (previous five trading days) and lagged cumulative abnormal returns (previous five trading days and the 25 days before that). Standard errors in parentheses are double clustered by firm and by date via a block bootstrap procedure following [Thompson \(2011\)](#).

Table 4

How common is the social signal across platforms? *Heterogeneity in first principal component (PC1) by firm size.*

Panel A: PC1 of attention signals by firm size						
	Small			Medium		Large
StockTwits	0.577			0.620		0.566
Twitter	0.653			0.652		0.615
Seeking Alpha	0.490			0.436		0.549
Fraction of variation	49.5%			58.9%		72.5%
	(1.225)			(1.009)		(10.168)
Panel B: PC1 of sentiment signals by firm size						
	Small			Medium		Large
StockTwits	0.637			0.659		0.643
Twitter	0.661			0.665		0.670
Seeking Alpha	0.397			0.352		0.372
Fraction of variation	36.3%			40.1%		42.4%
	(0.129)			(0.258)		(0.288)
Panel C: PC1 of residualized attention signals by firm size						
	Residualize news			Residualize news & firm FEs		
	Small	Medium	Large	Small	Medium	Large
StockTwits	0.634	0.664	0.577	0.647	0.671	0.609
Twitter	0.668	0.675	0.618	0.672	0.681	0.627
Seeking Alpha	0.390	0.321	0.534	0.360	0.293	0.485
Fraction of variation	46%	54.4%	69.4%	45.7%	54.4%	68.1%
	(1.167)	(1.047)	(9.060)	(1.139)	(0.798)	(7.651)
Panel D: PC1 of residualized sentiment signals by firm size						
	Residualize news			Residualize news & firm FE		
	Small	Medium	Large	Small	Medium	Large
StockTwits	0.656	0.671	0.656	0.669	0.673	0.659
Twitter	0.674	0.678	0.682	0.679	0.680	0.680
Seeking Alpha	0.338	0.299	0.323	0.300	0.291	0.320
Fraction of variation	36%	39.6%	41.7%	36.2%	39.4%	41.2%
	(0.122)	(0.274)	(0.274)	(0.110)	(0.248)	(0.210)

Note: This table reports heterogeneity in the first principal components from Table 2 panel C and Table 3 columns 1–6. The sample is split into three groups by firm size: “small” refers to stocks whose market capitalization is below 2 billion; “medium” those between 2 and 10 billion; “large” those above 10 billion. Standard errors in parentheses are double clustered by firm and by date via a block bootstrap procedure following Thompson (2011).

Table 5

How common is the social signal across user types? *Evidence from users groups on StockTwits.*

Panel A: Correlations within StockTwits					
	Top 1%	Professional	Intermediate	Novice	No label
StockTwits attention	0.805	0.860	0.948	0.914	0.947
StockTwits sentiment	0.095	0.108	0.118	0.088	0.166
Panel B: PC1 of residualized attention signals					
	PC1	PC2	PC3	PC4	PC5
Top 1%	0.415	0.800	0.432	0.003	−0.023
Professional	0.443	0.195	−0.783	0.372	0.121
Intermediate	0.466	−0.193	−0.124	−0.490	−0.700
Novice	0.445	−0.464	0.423	0.632	−0.094
No label	0.465	−0.263	0.081	−0.471	0.698
Fraction of variation	84.3%	7.2%	4.6%	2.3%	1.5%
	(0.077)	(1.131)	(0.863)	(0.405)	(0.276)
Panel C: PC1 of residualized sentiment signals					
	PC1	PC2	PC3	PC4	PC5
Top 1%	0.572	−0.182	−0.128	0.084	−0.785
Professional	0.472	−0.398	−0.473	0.311	0.547
Intermediate	0.381	0.028	0.819	0.389	0.179
Novice	0.281	0.896	−0.271	0.201	0.062
No label	0.475	0.062	0.126	−0.839	0.222
Fraction of variation	27.6%	19.5%	19%	17.9%	16%
	(0.079)	(0.027)	(0.031)	(0.043)	(0.060)

Note: This table reports the correlations and principal component analyses of social signals across different user types on StockTwits. Panel A reports the bivariate correlations of attention and sentiment between StockTwits signals from each user group and their complements. Panels B and C use residualized attention and sentiment signals, respectively, i.e., residuals from regressing each signal on DJNW sentiment (lagged 0 through 7 days), DJNW attention (lagged 0 through 7 days), dummies for earnings announcements (lagged 0 through 7 days), dummies for 5-k filings (lagged 0 through 7 days), and firm fixed effects. PCA of non-residualized social signals from StockTwits user subgroups are reported in Appendix Table A1. Standard errors in parentheses are double clustered by firm and by date via a block bootstrap procedure following Thompson (2011).

5.3 Table 4 and Table 5: firm-size and user-type heterogeneity

Read:

- In Table 4, attention commonality is much stronger for large firms than for small firms: 72.5% versus 49.5% of variance explained by PC1.
- Sentiment commonality also rises with size, but only modestly: 42.4% versus 36.3%.
- In Table 5, attention across StockTwits user groups is extremely common: PC1 explains 84.3%.
- Sentiment across the same user groups is much less common: PC1 explains 27.6%.

Economic message. Platform differences in sentiment are partly differences in *who posts*. The strongest fragmentation shows up exactly where user heterogeneity should matter most.

5.4 Table 6 and Figure 3: why sentiment and attention predict returns differently

Table 6
How do next-day returns relate to social signals?

	Dependent var.: $AR_{i,t+1}$ (%)				
	(1)	(2)	(3)	(4)	(5)
Sentiment $PC1_{i,t}(z)$	0.061*** (0.009)				
Attention $PC1_{i,t}(z)$	-0.131*** (0.050)				
StockTwits sentiment $_{i,t}(z)$		0.049*** (0.008)	0.053*** (0.008)		
StockTwits attention $_{i,t}(z)$		-0.206*** (0.056)	-0.151*** (0.051)		
Twitter sentiment $_{i,t}(z)$		0.011 (0.007)		0.022*** (0.007)	
Twitter attention $_{i,t}(z)$		0.112*** (0.031)		-0.010 (0.020)	
Seeking Alpha sentiment $_{i,t}(z)$		0.079*** (0.010)			0.081*** (0.010)
Seeking Alpha attention $_{i,t}(z)$		-0.010 (0.010)			-0.015 (0.010)
DJNW sentiment $_{i,t}(z)$	0.078*** (0.008)	0.067*** (0.008)	0.079*** (0.008)	0.082*** (0.008)	0.071*** (0.008)
DJNW attention $_{i,t}(z)$	0.012 (0.011)	-0.019* (0.010)	0.008 (0.010)	-0.008 (0.010)	-0.012 (0.009)
8-K report day $_{i,t}$	0.070 (0.044)	0.014 (0.044)	0.061 (0.043)	0.044 (0.044)	0.037 (0.042)
EA day $_{i,t}$	-0.547*** (0.091)	-0.539*** (0.091)	-0.547*** (0.091)	-0.538*** (0.091)	-0.566*** (0.091)
Volatility $_{i,t-5:t-1}$	-0.045 (0.375)	0.022 (0.378)	-0.009 (0.377)	-0.099 (0.375)	-0.114 (0.373)
CAR $_{i,t-30:t-1}$	-0.007*** (0.002)	-0.007*** (0.002)	-0.007*** (0.002)	-0.008*** (0.002)	-0.008*** (0.002)
CAR $_{i,t-30:t-1}$	-0.001* (0.001)	-0.001* (0.001)	-0.001* (0.001)	-0.001* (0.001)	-0.001* (0.001)
Log Google ASV $_{i,t}(z)$	-0.053*** (0.017)	-0.054*** (0.017)	-0.050*** (0.017)	-0.064*** (0.017)	-0.065*** (0.017)
Sentiment & Attention $_{i,t-5:t-1}$	Y	Y	Y	Y	Y
Firm (i) FE	Y	Y	Y	Y	Y
Date (t) FE	Y	Y	Y	Y	Y
Outcome mean	-0.048	-0.048	-0.048	-0.048	-0.048
Outcome SD	7.124	7.124	7.124	7.124	7.124
Observations	819,210	819,210	819,210	819,210	819,210
R ²	0.0320	0.0323	0.0321	0.0318	0.0319

Note: This table reports how next-day abnormal returns relate to social signals. The outcome is each security's abnormal return (AR) on day $t+1$ in percentage points. We control for DJNW standardized sentiment and attention, 8-K report date indicators, earnings announcement indicators, lagged return volatility (previous five trading days), lagged cumulative abnormal returns (previous five trading days and the 25 days before that), and Log Google ASV. All regressions include ten lags ($t-1$ to $t-10$) of sentiment, and ten lags of attention. (z) denotes a standardized variable (mean 0, standard deviation 1 using the estimation sample statistics). PC1 is the first principal component of StockTwits subsignals (top 1%, self labeled, professional, intermediate, novice, no experience), Twitter, and Seeking Alpha. Standard errors in parentheses are double clustered by firm and by date. *** 1%, ** 5%, * 10% significance level.

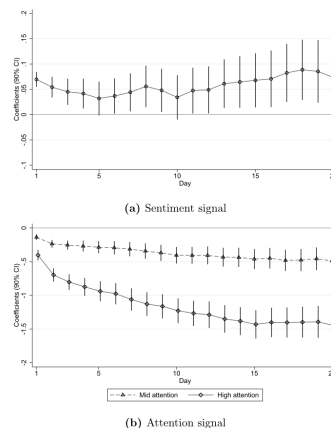


Fig. 3. Informationness of social signals for cumulative abnormal returns. Note: This figure plots the estimated coefficients (and 90% confidence intervals) on the first principal component (PC1) of sentiment signal (a) and on high-level and mid-level attention in Panel (b). High-level (or mid-level) attention equals 1 if the firm-day attention PC1 is above the 67th percentile (or between percentiles 33 and 67) within a year, and zero otherwise. The outcome is the cumulative abnormal return as of day $t+1, t+2, \dots, t+20$, starting on day $t+1$. We estimate separate regression for each horizon, and sentiment and attention coefficients are estimated in the same regression. (z) denotes a standardized variable (mean 0, standard deviation 1 using estimation sample statistics). Everything else follows Table 6 column 1.

Read:

- Table 6 shows that the common sentiment component predicts positive next-day returns: about +6.1 bps per one-standard-deviation increase.
- The common attention component predicts negative next-day returns: about -13.1 bps.
- StockTwits alone is close to those common-component estimates, meaning it is the platform that best summarizes the overall social signal.
- Figure 3 shows that sentiment's positive return effect remains positive through day $t+20$, with no meaningful reversal.
- High-attention days, in contrast, show a progressively more negative cumulative return over the next 10-20 days.

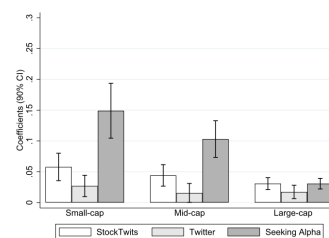
Economic message. Sentiment looks informative; attention looks like overreaction and later reversal. The paper’s key contribution is not only that both matter, but that they matter in *opposite ways*.

5.5 Table 7 and Figure 4: same-day buying pressure and where the attention effect lives

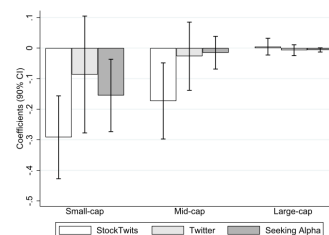
Table 7
How do same-day returns and retail trading relate to social signals?

	Dependent var.: AR _t (%)		RT imbalance _t (%)		RH user ratio _t (%)	
	(1)	(2)	(3)	(4)	(5)	(6)
Sentiment PCI _{t-1,t}	1.591*** (0.086)	1.498*** (0.033)	0.781*** (0.046)	0.655*** (0.043)	1.373 (1.025)	1.182 (1.055)
Attention PCI _{t-1,t}	3.630*** (0.734)		1.684*** (0.214)	7.053*** (1.701)		
Mid attention _t		1.496*** (0.058)		2.000*** (0.089)		1.897*** (0.531)
High attention _t		4.049*** (0.155)		3.969*** (0.143)		5.976*** (1.337)
DJNW sentiment _{t-1,t}	0.425*** (0.023)	0.380*** (0.021)	0.135*** (0.025)	0.073*** (0.024)	0.031 (0.073)	-0.029 (0.074)
DJNW attention _{t-1,t}	-0.272*** (0.046)	0.019 (0.036)	-0.134*** (0.044)	-0.107** (0.042)	-0.303 (0.358)	0.303 (0.260)
Sentiment & Attention _{t-1,t,...,t-10}	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y
Firm (i) FE	Y	Y	Y	Y	Y	Y
Date (i) FE	Y	Y	Y	Y	Y	Y
Outcome mean	0.358	0.358	-0.327	-0.327	3.705	3.705
Outcome SD	9.190	9.190	23.173	23.173	178.715	178.715
Observations	815,706	815,706	810,652	810,652	171,479	171,479
R ²	0.0917	0.0779	0.0279	0.0200	0.1988	0.1989

Note: This table reports how same-day abnormal returns and retail trading relate to social signals. The outcome in columns 1–2, 3–4, and 5–6 is abnormal returns (AR) on day t in percentage points, retail trading (RT) imbalance on day t in percentage points, and Robins Hood (RH) user ratio on day t in percentage points, respectively. RH user ratio on day t is calculated as (user number around 4 pm on t /number around 4 pm on $t-1$) - 1 following Barber et al. (2012). High (or mid) attention equals 1 if the firm-day attention PCI is above 67 percentile (or between 33 and 67 percentile) within a year, and zero otherwise. We control for DJNW standardized sentiment and attention, 8-K report date indicators, earnings announcement indicators, lagged return volatility (previous five trading days), lagged cumulative abnormal returns (previous five trading days and the 25 days before that), and log Google ADV. We additionally control for lagged RT imbalance (previous five trading days and the 25 days before that) in columns 3–4 and lagged RH user ratio (previous five trading days and the 25 days before that) in columns 5–6. All regressions include ten lags ($t-1$ to $t-10$) of sentiment, and ten lags of high and mid attention indicators. (i) denotes a standardized variable (mean 0, standard deviation 1 using the estimation sample statistics). PCI is the first principal component of StockTwits subsignals (top 1%, self labeled, professional, intermediate, novice, no experience), Twitter, and Seeking Alpha. Standard errors in parentheses are double clustered by firm and by date. *** 1%, ** 5%, * 10% significance level.



(a) Informativeness of sentiment signal



(b) Informativeness of attention signal

Fig. 4. How do next-day returns relate to platform-specific social signals by firm size? Note: This figure plots the estimated coefficients (and 90% confidence intervals) on sentiment and attention signals (z) for StockTwits, Twitter, and Seeking Alpha for small-cap, mid-cap, and large-cap firms, separately. Firm size categories follow those in Table 1. The outcome is the abnormal return on day $t+1$. (z) denotes a standardized variable (mean 0, standard deviation 1 using estimation sample statistics by firm size). Everything else follows Table 6 columns 3–5.

Read:

- Table 7 shows that same-day abnormal returns and same-day retail buying are strongly positive on high-attention days.
- The same-day relation is especially strong for attention, which is exactly what one would expect if attention partly proxies for retail demand pressure.
- Figure 4 shows that the negative next-day attention effect is strongest for small- and mid-cap firms and close to zero for large-cap firms.

Economic message. The attention story is most naturally a retail-overreaction story, and that story is strongest where social media and retail trading matter most: smaller firms.

5.6 Table 8 and Figures 5–6: the StockTwits character-limit change

Read:

- Figure 5 shows that only the upper tail of the message-length distribution visibly expands after May 8, 2019.
- Table 8 shows that StockTwits sentiment becomes more informative after the change: the post $\times$ sentiment coefficient is about **+0.070%**, and rises to about **+0.138%** for stocks in the top quartile of message length.

sentiment coefficient for the common PC signal is about -0.125% , roughly offsetting the pre-event positive coefficient of about $+0.111\%$.

- That decline is concentrated among *new* StockTwits users: the post $\times$ sentiment coefficient is about -0.103% for new users and approximately zero for old users.
- Figure 7 shows a dramatic jump in “short squeeze” discussion by new users after January 2021, with a much smaller shift among established users.

Economic message. The quality of the sentiment signal depends on who is posting. After the GME episode, new-user narratives appear to dilute the predictive content of social-media sentiment.

6 Short summary

- The paper compares investor sentiment and attention across Twitter, StockTwits, and Seeking Alpha.
- Attention is strongly common across platforms; sentiment is weakly common and much more platform- and user-specific.
- These results survive controls for traditional news, disclosures, firm fixed effects, and lagged return variables.
- The common sentiment component predicts positive next-day returns and shows no reversal, which is consistent with information.
- The common attention component predicts negative next-day returns, after a large positive same-day move and retail buying pressure, which is consistent with overreaction and partial reversal.
- StockTwits is the platform whose raw signal most closely lines up with the common social signal.
- The 2019 StockTwits character-limit increase makes StockTwits sentiment more informative, especially for long messages.
- The 2021 GME episode makes sentiment much less informative, and the decline is concentrated among new StockTwits users.
- The paper’s big message is that “social media” is not one signal: attention and sentiment differ, platforms differ, and user composition and platform design materially shape what these signals mean for returns.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes three core facts.

- Across the major investor social platforms, **attention is common** and **sentiment is fragmented**.

- These two components have **different return signatures**: sentiment looks informative, attention looks like overreaction and reversal.
- Both **platform design** and **user composition** materially affect what a platform-level signal contains.

7.2 Economic intuition

A natural way to summarize the economics is this:

- Attention tracks collective focus. When many investors talk about the same stock, that signal spreads broadly across platforms and tends to coincide with retail buying pressure.
- Sentiment is closer to a communication product: it depends on who is speaking, how much space they have, what norms the platform creates, and how users interpret or frame ideas.
- Because of that, sentiment is less portable across platforms than attention.

The event studies fit this intuition closely. More room to explain an idea makes StockTwits sentiment more informative; a surge of new speculative users focused on short squeezes makes the sentiment signal less informative.

7.3 Takeaway

For empirical work, the paper's lesson is straightforward: one should not pool all investor social-media data into a single black box. The choice of platform, the distinction between sentiment and attention, and the composition of users all matter for measurement and for interpretation. For finance more broadly, the paper provides a microfoundation for why attention booms can coexist with fragile or deteriorating sentiment quality during speculative episodes.